

String Theory Is Complexity-Disfavored: A Description-Length Bound

Gary Abraham Bernstein

Independent Researcher ORCID: 0009-0009-1761-2867

Abstract

String theory is exponentially disfavored under every standard parsimony measure. It requires 3,000 to 4,700 bits of specification to reproduce observed physics. The Standard Model requires roughly 60 bits at its minimum description length, bounded by the output type of a finite scalar tuple. The prior odds ratio exceeds $2^{2,940}$. Under algorithmic probability, Minimum Description Length, or Bayesian model selection, this gap is decisive. The conclusion is code-relative and conditional. It states a prior penalty, not a refutation.

Keywords: Kolmogorov complexity, algorithmic probability, minimum description length, description-length bounds

1. Introduction

How many bits does it take to specify a theory of physics? Under algorithmic probability (Solomonoff, 1964), every additional bit of specification halves a theory's prior probability. A theory requiring 10 more bits than a competitor is 2^{10} , roughly 1,000, times less probable. A theory requiring 100 more bits is 2^{100} times less probable, a number larger than the count of atoms in the universe.

The Standard Model is a finite list of scalar parameters. Under any compressing rule, its specification cost is on the order of 60 to 100 bits. String theory routes through a landscape of at least 10^{500} vacuum solutions, each requiring a compactification geometry, flux integers, and stabilized moduli. Its specification cost is 3,000 to 4,700 bits. The difference is at least 2,940 bits. The prior probability ratio is $2^{2,940}$. String theory is not slightly disfavored. It is exponentially, absurdly, disfavored.

This paper makes that comparison precise.

2. What “bits of specification” means

Three levels of description-length reasoning appear in this paper. They are related but distinct.

Kolmogorov complexity $K(x)$ is the length of the shortest program that generates x on a universal Turing machine. It is uncomputable. No numerical estimate in this paper claims to compute exact K .

Explicit code length is the length of a specific prefix code that produces x . Any explicit code length is an upper bound on K . This is what the paper measures. When we say “1,661 bits for landscape selection,” we mean a self-delimiting index into a set of 10^{500} requires at least that many bits. The true K could be lower if a short generating rule exists. None has been found.

Algorithmic probability weights structures by $P(S)$ proportional to $2^{(-K(S))}$. Shorter descriptions get exponentially more weight. Solomonoff (1964) showed this prior dominates all computable alternatives. Rissanen’s (1978) Minimum Description Length and Bayesian model selection with proper priors all penalize description length in the same direction.

Toy example

Two models explain a binary sequence of length n . Model A: a single bias parameter. Model B: a lookup table listing each bit. Model A’s description length is roughly $\log(1/\text{precision})$. Model B’s is n bits. For large n , every parsimony measure favors Model A. The string theory comparison has the same structure at vastly larger scale.

3. The Standard Model: a short description

The Standard Model contains 19 to 26 free parameters: quark masses, lepton masses, coupling constants, mixing angles, the Higgs mass, and the QCD vacuum angle. Its output type is a finite tuple of real scalars.

Whether these constants admit a short generating rule is unknown. If they do, K could be very small, on the order of 60 to 100 bits. If they are algorithmically random at measured precision, K scales with precision. Either way, the output type is a finite list of numbers. No landscape selection. No geometric specification. No flux assignment. The description is short because the output is simple.

4. String theory: a long description

String-theoretic realizations route through a higher-level construction. Each component adds bits:

Landscape selection. The string landscape contains approximately 10^{500} distinct vacuum solutions. Specifying which one reproduces our physics requires $\log_2(10^{500})$, approximately 1,661 bits. More recent F-theory estimates (Taylor & Wang, 2015) suggest $10^{272,000}$ consistent flux vacua. Larger landscapes mean more bits.

Compactification geometry. Six extra dimensions compactify to a Calabi-Yau threefold. Specifying one from the Kreuzer-Skarke database of roughly 473 million candidates requires approximately 29 bits for the Hodge pair alone. Additional topological and geometric data add more.

Flux integers. Flux values on approximately 500 topological cycles, with typical ranges of 10 to 100 per cycle, contribute approximately 1,000 to 2,500 bits.

Moduli stabilization. Approximately 100 complex moduli must be stabilized at specific values, contributing approximately 200 to 500 bits.

Total: 3,000 to 4,700 bits. Each component is an upper bound on the specification cost of that component. The total is conservative.

5. The comparison

A program generating a list of numbers has lower minimum K than a program constructing a specific complex geometry from 10^{500} options. The gap persists at every level of compression. String theory’s “explanation” has higher Kolmogorov complexity than what it purports to explain. Under any information-theoretic parsimony principle, this is the opposite of scientific progress.

Proposition. Let L_{SM} and L_{ST} be the explicit description lengths of the Standard Model and a string-theoretic realization under the same prefix code. Under the universal prior:

$$P(ST) / P(SM) \approx 2^{-(L_{ST} - L_{SM} + c)}$$

where c is a small constant. For $L_{ST} - L_{SM}$ approximately 2,940 bits, the ratio is at most $2^{-(2,940)}$.

6. Objections

“A short selection rule might exist.” If someone finds a compact rule that picks the right vacuum from 10^{500} without explicit enumeration, the selector cost collapses. This is the escape hatch. No such rule has been found. The burden of exhibiting one falls on string theory’s proponents.

“Vafa’s Swampland program reduces the landscape.” If the consistent landscape shrinks to 10^{100} , the selector cost drops from 1,661 bits to approximately 332 bits. Total cost reduces to roughly 2,000 bits. The gap narrows. It does not close.

“Algorithmic probability is not the right measure.” The argument does not depend on Solomonoff specifically. Any formal parsimony principle that penalizes description length, MDL, Bayesian model selection, or any computable prior, yields the same qualitative conclusion. The specific ratio differs. The direction does not.

“The Standard Model has unexplained parameters too.” Yes. Its K is not zero. But its output type is a finite scalar tuple. String theory’s output type requires landscape selection, geometric specification, and flux assignment. The output types have different complexity floors. That is the comparison.

“This does not refute string theory.” Correct. Under mathematical monism, all consistent structures exist. String theory describes a consistent structure. The claim is that it describes a lower-measure one: our universe is exponentially more likely to be one of the simpler structures that reproduce the same physics.

7. Alternative candidates

The paper is not about which theory is right. It is about which theory is simpler. For reference:

The most algorithmically favored candidates are those that collapse framework, parameters, and dynamics into a single short rule. Wolfram-style hypergraph programs, if one is found that reproduces known physics, would have specification costs below 100 bits. The Standard Model, despite its unexplained parameters, remains the information-theoretically favored description of known physics.

The theory that explains the most with the least specification wins. String theory increases specification complexity. That is the problem.

8. Conclusion

String theory requires 3,000 to 4,700 bits to specify a realization that reproduces observed physics. The Standard Model requires roughly 60 to 100. The ratio is $2^{2,940}$ or larger. Under any parsimony measure that penalizes description length, string theory is exponentially disfavored. The conclusion is conditional on the coding scheme and does not bear on mathematical consistency. It states a prior penalty, not a refutation.

The escape is to produce a short generating rule that replaces the explicit landscape selector. Until that rule is exhibited, the specification cost stands.

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